NI TB Genomic Surveillance

Outbreak Investigation Report

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About This Report

This report is produced by the Northern Ireland TB Genomic Surveillance platform using whole-genome sequencing (WGS) data and epidemiological case records. It is intended to support TB programme staff and public health investigators by providing genomic evidence for transmission clusters, drug-resistance profiles, and programme performance metrics. **This is a decision-support tool only — all findings must be reviewed and acted on by a qualified clinician or public health professional. No automated decisions are made.**

TB Genomics — Key Concepts

Concept	Explanation
Whole-Genome Sequencing (WGS)	Reads the complete ~4.4 Mb genome of M. tuberculosis. More informative than conventional typing (MIRU, spoligot
SNP (single nucleotide polymorphism)	A single base-pair difference in the genome. Closely related strains share very few SNPs. Used as a genetic 'distan
SNP threshold for transmission	Strains with ≤12 SNPs are considered potentially linked (UK NICE guidance). ≤5 SNPs suggests recent direct trans
Lineage	M. tuberculosis is classified into 7+ major lineages (L1–L7) reflecting global evolutionary history. Lineage influences
Cluster	A group of cases whose sequences are genetically similar (within the SNP threshold). A cluster does not prove dire
outbreaker2	A Bayesian MCMC method that combines SNP distances with collection dates and an assumed generation time to
Generation time	The average time between one person being infected and the next person they infect being detected. For TB, this is
MCMC convergence	Markov Chain Monte Carlo simulations must reach a stable state ('converge'). A convergence diagnostic near 1.0 in
Drug resistance	Genomic mutations predict resistance to first-line drugs (isoniazid, rifampicin, etc.) and define MDR-TB (multi-drug r
Transmission network	A directed graph where arrows indicate the most probable direction of transmission. High-confidence links (posterio

Table 1. TB genomics reference — key terms used throughout this report.

Case Summary

Total Cases	250
Clustered Cases	131
Unclustered Cases	119
Open Clusters	6

Table 2. Programme case count summary. Clustered cases are those linked by genomic similarity to at least one other case. Unclustered (singleton) cases may represent imported strains, sporadic transmission, or reactivation of latent disease. Open clusters are active genomic transmission clusters with ongoing epidemiological investigation.

Automated Interpretation Flags

- Open clusters requiring investigation: 6.
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	1500
MCMC Iterations	2000

Burn-in	500
Mean Log-Likelihood	-176103.83
Log-Likelihood SD	30617.42

Table 3. outbreaker2 Bayesian MCMC analysis parameters. **Posterior Samples** is the number of accepted MCMC draws used to compute estimates — higher values give more stable posteriors. **Mean Log-Likelihood** reflects model fit; values closer to zero (less negative) indicate better fit. **Transmission Probability** is the average posterior probability that any given case-pair represents a direct transmission event. **Generation Time** is the modelled average interval (in days) between infection events in a transmission chain. **Convergence Diagnostic** near 1.0 confirms the MCMC chain has stabilised; values >1.1 indicate cautious interpretation is needed.

How to interpret outbreaker2 results: outbreaker2 reconstructs the most probable transmission tree using both genetic distance (SNPs) and timing (collection dates). Pairs with high posterior transmission probability (>0.5) represent genomically and temporally plausible direct transmission events. These are candidates for epidemiological investigation to identify shared exposure. Lower probability pairs may still be linked within the same cluster but through one or more undetected intermediate cases.

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	32
Sequenced Cases	27
Sequencing Coverage (%)	84.38
QC Reported Cases	0
QC Pass Cases	0
QC Fail Cases	0
QC Pass Rate (%)	None
Contamination Flags	0
Median Days Specimen to QC	None

Table 4. Programme surveillance KPIs over the reporting window. **Sequencing Coverage** is the percentage of eligible TB culture-confirmed cases that have received whole-genome sequencing. The UK target is ≥80%. **QC Pass Rate** is the percentage of sequenced samples that meet quality thresholds (e.g. ≥95% genome coverage at ≥10×). Low pass rates may indicate DNA quality issues, contamination, or laboratory process variation. **Contamination Flags** indicate samples where a mixed-strain signal suggests cross-contamination requiring repeat or rejection.

Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
United Kingdom	32	27	84.38

Table 5. Regional sequencing representativeness. Regions with coverage <80% may introduce ascertainment bias — clusters in under-sequenced regions may be underdetected. Where persistent regional gaps exist, review laboratory submission pathways and specimen transport processes.

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-16	1	1	100.00	None
2026-02-23	1	1	100.00	None
2026-03-02	3	3	100.00	None
2026-03-09	5	4	80.00	None
2026-03-16	4	3	75.00	None
2026-03-23	3	3	100.00	None
2026-03-30	1	0	0.00	None
2026-04-06	2	2	100.00	None

2026-04-13	5	4	80.00	None
2026-04-20	0	0	None	None
2026-04-27	2	2	100.00	None
2026-05-04	5	4	80.00	None

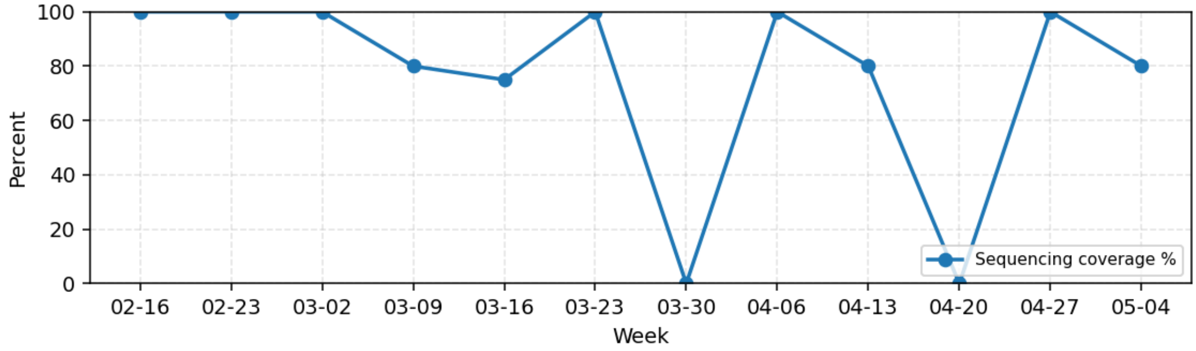

Table 6. Weekly sequencing coverage and QC pass rates over the last 12 weeks. Figure 1 (below) plots coverage and QC trends — a declining trend may indicate emerging laboratory issues. Weeks with zero eligible cases may reflect reporting lags rather than true absence of TB.

Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
0cd6321d	27	1	2026-04-13	open	62
8a0a5195	25	1	2026-03-28	open	57
0fb1ba30	24	1	2026-05-02	open	57
b9bdab14	21	1	2026-05-08	open	51
941d8d04	20	1	2026-05-05	open	49
218d446f	14	1	2026-05-09	open	37

Table 7. Clusters ranked by investigation priority score. Score is composite: cluster size (×2), cross-region spread (×3), specimen recency within 14/30/60 days (×3/2/1), open investigation status (×3). Higher scores indicate clusters warranting urgent epidemiological follow-up. **Status 'open'** means an active field investigation is ongoing or recommended.

Recommended action: For clusters with priority score >10 and status 'open', ensure field epidemiology is actively investigating shared exposure (household contacts, healthcare settings, social networks). Cross-region clusters may indicate transmission events during travel or care-seeking across NHS trust boundaries.

Lineage and Drug Resistance Validation

Overall Status	completed
TB-Profiler	installed_but_unusable
Mykrobe	not_installed
Docker Fallback	True
Docker Daemon Running	False
FASTA Inputs	1

Table 8. Lineage and drug-resistance validation status. TB-Profiler and Mykrobe are bioinformatic pipelines that classify M. tuberculosis lineage and predict drug resistance from WGS reads. 'Available' means the tool executed successfully; 'unavailable' may indicate missing software, Docker daemon issues, or insufficient FASTA inputs. FASTA inputs refers to the number of consensus genome sequences submitted for analysis.

Lineage/DR Next Steps

- Install tb-profiler/mykrobe dependencies (WSL2 micromamba env is supported via TBPROFILER_WSL_FALLBACK=1).
- If local dependencies fail, install Docker Desktop and rerun with TBPROFILER_DOCKER_FALLBACK=1.
- Place FASTQ/VCF/FASTA inputs in uploads/ (or export pipeline outputs there).
- Optionally provide exports/lineage_resistance_calls.csv to hydrate tb_interpretation.

Secondary Transmission Engines

Secondary Validation Status	completed
Consensus State	pending_secondary_outputs
TransPhylo	not_installed
BactDating	not_installed
Tree Input Available	False
Cases Input Available	True

Table 9. Secondary engine validation. TransPhylo uses a phylogenetic tree and sampling dates to reconstruct transmission under a within-host evolutionary model. BactDating estimates dated ancestral phylogenies to calibrate transmission timelines. Both require a Newick-format phylogenetic tree as input. Where tools are unavailable, outbreaker2 results remain the primary genomic evidence.

Cross-Method Clustering Comparison

Sequence Assigned Cases	131
Outbreaker Assigned Cases	250
Overlap Cases	131
Pairwise Precision	0.171
Pairwise Recall	0.937
Pairwise Jaccard	0.169

Table 10. Agreement between SNP-threshold sequence clustering and outbreaker2 probabilistic clustering. **Precision**: of pairs grouped together by outbreaker2, the fraction also grouped by sequence clusters. **Recall**: of pairs grouped by sequence clusters, the fraction also grouped by outbreaker2. **Jaccard**: overall overlap index (0=no agreement, 1=perfect agreement). Discordant pairs — grouped by one method but not the other — may represent cases where temporal data (outbreaker2) overrides genomic distance alone, or where the SNP threshold is set differently.

Sequence Clustering Snapshot

Cluster Count	6

Transmission Priority Signals

Understanding TB Transmission Routes from WGS

Whole-genome sequencing identifies genomic relatedness but does not directly observe contact events. Combining genomic clusters with epidemiological data (contact tracing, shared locations, timeline of diagnosis) allows investigators to build a plausible transmission chain. The table below summarises the genomic signals and their transmission implications.

Genomic Signal	SNP Range	Transmission Implication	Recommended Action
Highly probable direct transmission	0–5 SNPs	Strong genomic evidence of recent direct person-to-person transmission. Strain has had little time to evolve.	Immediate contact tracing; identify shared setting (household,
Possible direct or near-direct transmission	6–12 SNPs	Genetically close; consistent with transmission within the last 1–2 years or via an unidentified intermediate.	Epidemiological linkage investigation; check whether cases sh
Within extended cluster — indirect likely	13–50 SNPs	Genetically related but too diverged for recent direct transmission. Likely share a common ancestor strain.	Review cluster history; may represent reactivation from the sa
Unrelated strains	>50 SNPs	No plausible genomic link. Coincident diagnoses are likely due to independent exposures or reactivation.	No cluster-based action; manage as separate cases.
Mixed-lineage / contamination	N/A	Two or more distinct strain signals in a single sample. May indicate laboratory cross-contamination or mixed infection.	Flag for repeat sequencing; do not use in cluster assignment unt

Table 11. TB transmission route classification by SNP distance (M. tuberculosis whole-genome comparison). SNP thresholds follow UK NICE guideline NG33 and published literature (Walker et al. 2013, Meehan et al. 2019). The generation time assumed in outbreaker2 modelling is set at programme configuration and affects when a given SNP distance is interpreted as consistent with direct vs indirect transmission.

Transmission Priority Signals

Case	Region	Risk	Out	In
a0026426		7.945	11	1
9e385e98		7.4629	11	1
cb6dabf2		5.9	8	1
d39207e8		5.5896	8	1
33423c4b		5.2135	8	1
9d1257ed		3.8	5	1
5fd39b4e		3.8	5	1
b97a6131		3.681	5	1
259b7b08		3.6743	5	1
7c7c0b18		3.5755	5	1

Table 12. Top-priority cases by network centrality. Network snapshot: 250 nodes, 249 directed links, 185 high-confidence links (posterior >0.70). **Out** = outgoing transmission links (potential sources); **In** = incoming links (potential recipients). Cases with multiple outgoing high-confidence links ('superspreaders') should be prioritised for epidemiological investigation.

High-confidence transmission links (posterior probability >0.70) represent the strongest genomic evidence of direct transmission between a pair of cases. Investigators should review the contact history for all such pairs. Where epidemiological linkage can be confirmed, the direction of transmission (source → recipient) inferred by outbreaker2 can inform contact prioritisation for LTBI screening.

Diagnostic Graphics

The following plots are generated by outbreaker2 and the platform's supplementary visualisation pipeline. Each figure caption explains the content and how to interpret the output.

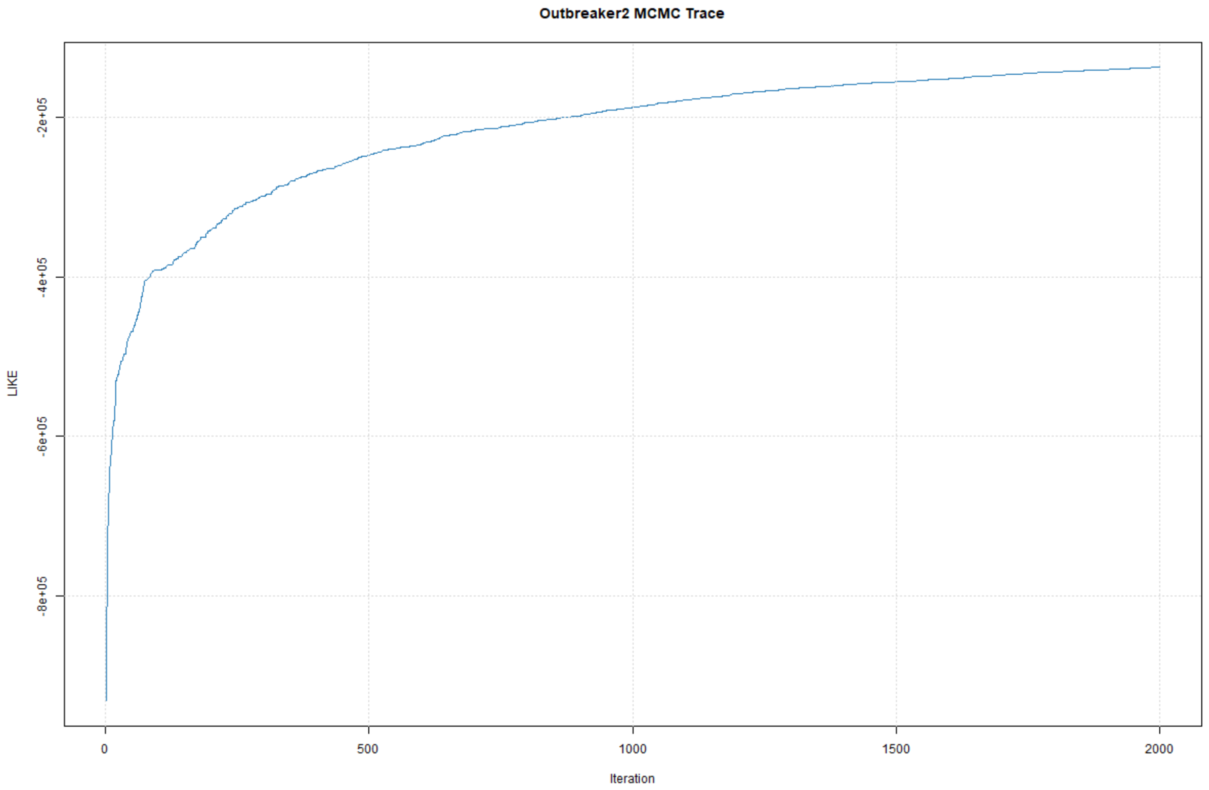

Figure 2. MCMC trace plot — log-posterior probability over iterations. A well-mixed chain shows stable fluctuation around a mean value (no upward/downward drift). If the trace shows a long burn-in slope or multiple plateaux, the chain may not have converged; consider increasing iterations or checking input data quality.

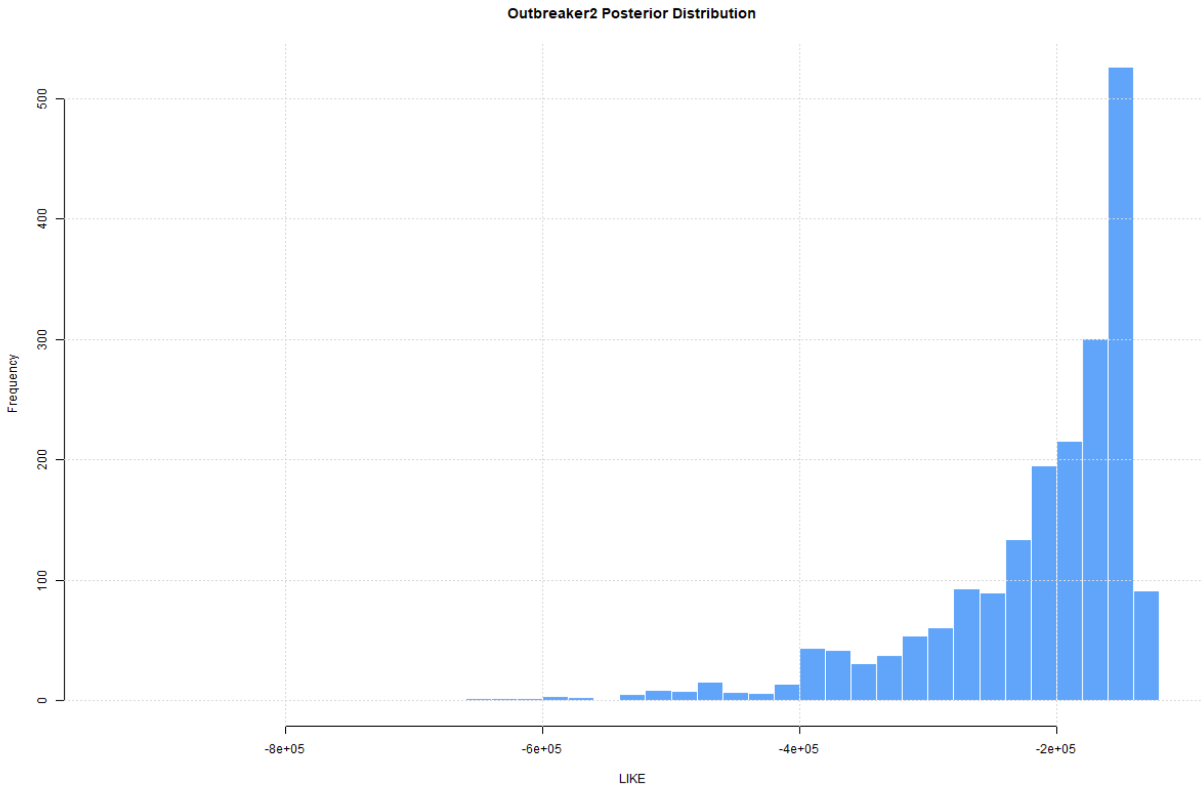

Figure 3. Posterior distribution histograms — marginal distributions of key model parameters (transmission probability, sampling probability, generation time). Narrow, symmetric peaks indicate well-determined parameters. Broad or multi-modal distributions suggest parameter uncertainty, which should be reflected in cautious interpretation of individual transmission links.

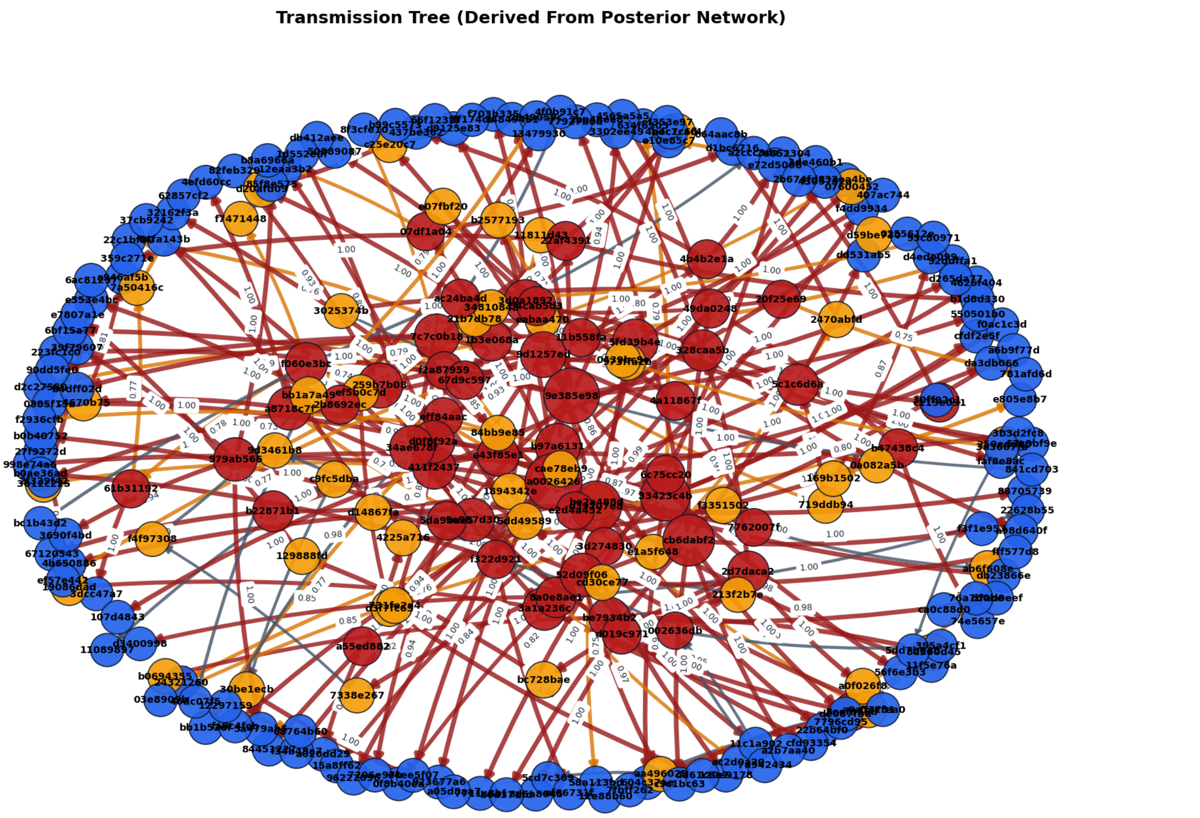

Figure 4. Inferred transmission tree (most probable who-infected-whom). Each node is a case; arrows indicate the direction of inferred transmission from source (tail) to recipient (head). Arrow thickness or colour (where shown) reflects posterior probability. Dashed or thin arrows

indicate lower-confidence links. Cases with no incoming arrow are probable index cases or represent undetected importation events.

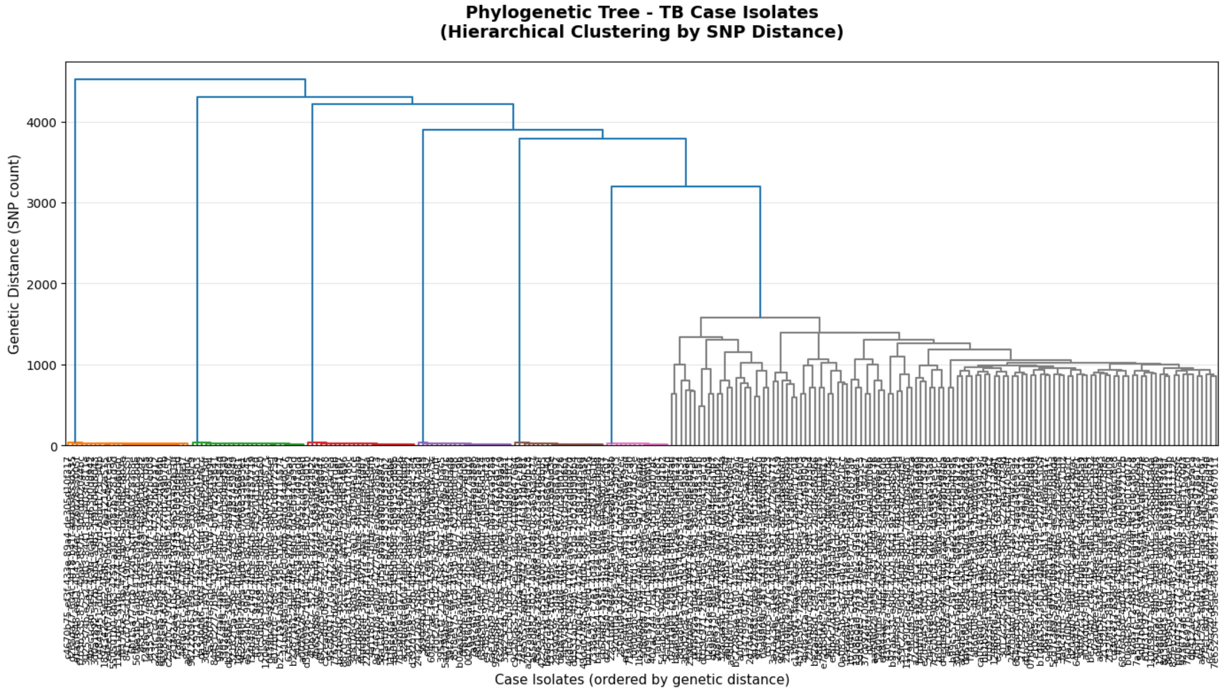

Figure 5. Phylogenetic context — a midpoint-rooted maximum parsimony or neighbour-joining tree of sequenced cases, coloured by cluster or region. Branch length represents SNP distance. Cases on short branches with few SNPs between them form tight clades consistent with recent transmission. Well-separated clades indicate genetically distinct strain lineages circulating concurrently.

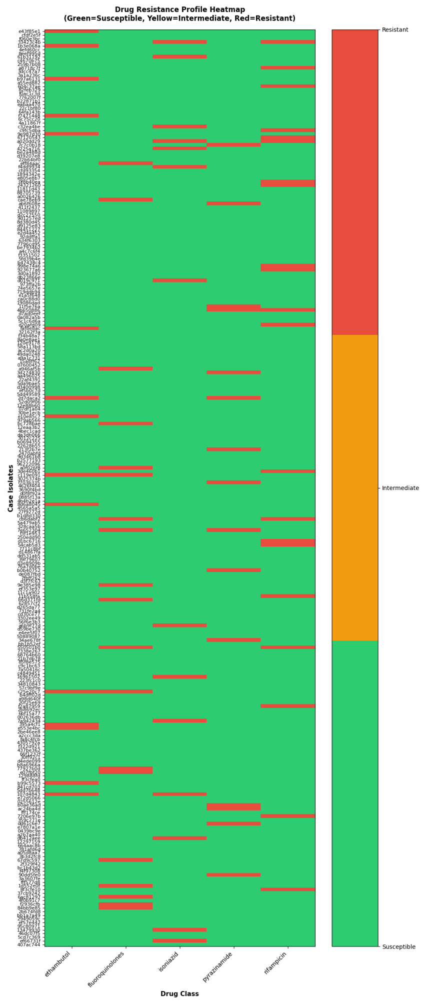

Figure 6. Drug resistance profile summary — frequency of predicted resistance mutations across the sequenced cohort. Bars represent the proportion of cases with predicted resistance to each antibiotic class. Rifampicin + isoniazid co-resistance defines MDR-TB. High frequencies of any first-line resistance warrant urgent review of empirical treatment protocols.

M. tuberculosis Lineage Reference

Lineage classification places a strain within the global phylogeny of M. tuberculosis. Lineage influences drug-resistance acquisition patterns and geographic origin. The following table provides clinical and epidemiological context for lineages commonly observed in Northern Ireland.

Lineage	Name / Origin	DR Association	NI Relevance	Notes
L1	East-African-Indian / Indo-Oceanic	Lower MDR-TB frequency	Cases linked to South Asia, Horn of Africa	Common; found in Bangladeshi, Indian, Somali communities.
L2	East-Asian (Beijing lineage)	High MDR/XDR-TB risk; associated with resistance acquisition	Sporadic importation; watch for Beijing strains	Beijing strains have shown high transmissibility in some outbre
L3	East-African-Indian (Delhi/CAS)	Moderate DR frequency	Cases linked to Pakistan, Afghanistan, India	Common in large urban TB programmes in the UK.
L4	Euro-American	Generally lower DR; but historical MDR clusters exist	Dominant UK-born strain type	Most legacy UK TB is L4. Reactivation common in older cohort
L5 / L6	West-African (Mycobacterium africanum)	Lower overall DR	Cases linked to West Africa	Slower growth, may present with atypical features.
L7	Ethiopian	Limited data	Rare in NI	Emerging lineage classification; limited clinical guidance availa

Table 13. M. tuberculosis lineage reference for clinical and epidemiological context. DR = drug resistance; MDR = multidrug-resistant; XDR = extensively drug-resistant. Lineage assignment should be combined with phenotypic DST and clinical judgment. Source: Coll et al. (2014) Nature Genetics; WHO Global TB Report 2023.

Clinical and Public Health Action Summary

The table below maps genomic findings to recommended clinical and public health actions. All actions must be confirmed by the responsible clinician and public health team.

Finding	Recommended Action	Urgency
New case links to an existing open cluster (≤12 SNPs)	Notify cluster lead; extend contact tracing to include new case contacts; review whether the cluster case has been ide	Within 5 working days
High-confidence transmission link identified (posterior ≥0.70)	Epidemiological review of both cases; document shared exposure if found; update cluster investigation record.	Within 5 working days
New cluster opened (≥2 cases genetically linked, no existing cluster)	Open investigation; notify public health; assign epidemiologist; initiate contact tracing for all cases.	Within 2 working days
MDR-TB predicted by genomics	Confirm with phenotypic DST; notify MDR-TB specialist centre; initiate enhanced infection control if hospitalised.	Immediately
Sequencing coverage <80% for a region	Review specimen submission and transport processes for that region; identify cases that did not receive WGS and arra	Monthly programme review
MCMC convergence diagnostic >1.1	Do not rely on posterior transmission probabilities from this run. Increase MCMC iterations and re-run outbreaker2. Che	Before using results
Contamination flag on a sample	Exclude sample from cluster assignments; arrange repeat sequencing from original culture. Investigate laboratory proce	Within 10 working days

Table 14. Recommended clinical and public health actions mapped to genomic findings. Urgency thresholds align with PHE/PHA TB operational guidance. All genomic findings must be reviewed in conjunction with clinical history, contact tracing records, and microbiological DST before action is taken.

Disclaimer: Genomic cluster assignments and transmission inferences are probabilistic estimates based on mathematical models. They supplement but do not replace epidemiological investigation. Do not use genomic evidence alone to assign legal or clinical liability for transmission.

Data Provenance

This report combines outbreaker outputs with surveillance KPIs, lineage/DR validation, secondary engine readiness, and cross-method clustering comparison artifacts available at generation time.
Outbreaker summary timestamp: 2026-05-10T11:51:36Z
Transmission network timestamp: 2026-05-10T11:51:36Z
Lineage/DR validation timestamp: 2026-05-10T11:45:44.589335+00:00
Secondary engine validation timestamp: 2026-05-08T22:40:58Z
Method comparison timestamp: 2026-05-10T11:52:18.445335+00:00